

ChiPi Technologies

Industry-Guided Project Presentation

Project Experience Initiative • Real Projects • Real Mentorship • Real Outcomes

Student Profile

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Project Overview

Executive Summary

This presentation delivers a comprehensive overview of a production-ready ****Inpatient Department (IPD) Admission and Registration Portal****. The ecosystem features a secure registration landing gateway tied alongside an extensive, multi-column client intake workspace. Key design targets focus on strict layout separation rules, full local state management continuity, and fully responsive CSS grid system splits.

Problem Statement

The Challenge Being Solved

- **Disjointed Patient Ingest Workflows:** Standard medical administration systems suffer from disconnected interfaces separating original records search from live admission screens.
- **Complex Structural Density:** Handling complex multi-input lines (such as matching custom Date/Time pickers, multi-option lists, and specific action button segments) can create misaligned fields.
- **Session & Local Sync Failures:** Patient context data often gets cleared or mixed up during browser updates and page transitions.

Business Impact

Who Benefits

- **Healthcare Staff:** Complete verification checks in seconds without dealing with confusing navigation patterns.
- **Hospital Admins:** Drastically lower patient ingestion drops caused by complex web portals.

Value Generated

- Reduces form submission mistakes by enforcing structured drop-down options and read-only field safety blocks.
- Improves clinical tracking speed by syncing application tokens straight across standard storage layers.

Requirements

Functional Requirements

- Safe fallback alerts for non-existent application strings.
- Persistent patient state context pass using global `sessionStorage` arrays.
- Multi-input arrays (Triple input filters, precise room options, attachment blocks).

Non-Functional Requirements

- Modular structural isolation splitting HTML views from `style.css` configurations.
- Fluid layouts using explicit responsive break boundaries (`@media (max-width: 1100px)`).
- No-clip container dimensions utilizing balanced box shadow rules.

Solution Architecture

System Design & Module Map

The system cleanly separates registration gateways from full admission worksheets while syncing global data states seamlessly:

Global Workspace Stylesheet (`style.css` definitions)

Login/Search Gateway (`index.html`)

Validates registration tokens via `localStorage` lookup models.

IPD Admission Console (`registration.html`)

2-Column intake system handling clinical context mapping.

UI/UX Design

Layout Aesthetics & Component Design

- **Balanced Forms Display:** Engineered a clean CSS grid layout split into left and right workspace blocks to prevent alignment issues across fields.
- **Clean Grouped Inputs:** Combined elements smoothly using custom CSS classes (``double-input``, ``triple-input``, ``age-box``) to match complex input fields perfectly.
- **Visual Hierarchy System:** Implemented active focus states (``:focus`` selectors) paired with orange interactive icons to draw focus to critical actions instantly.

Technology Stack

Tools & Technologies Applied

Frontend Design Frameworks:

HTML5 Forms

CSS3 Flexible Grids

Responsive Breakpoints

Pseudo-classes

Storage & State Processing APIs:

Web Storage API

localStorage Records

sessionStorage Cache

Core Engineering Competencies (BCA Context):

Document Object Model (DOM)

JSON Serialization

Dynamic Event Interceptions

Implementation Journey

Development Process Phases

1. **Data & Structure Mapping:** Laid out fields clearly inside HTML columns to support detailed patient intake data requirements.
2. **Advanced Style Mapping:** Implemented responsive flex and grid layouts to ensure form sections stack cleanly on narrower devices.
3. **Context Pipeline Wiring:** Connected lookup triggers to fetch and map saved session variables straight into matching input boxes on page load.

Challenges Faced

Obstacles Encountered

- **Complex Multi-Row Misalignments:** Managing uniform spacing across sections containing uneven elements (like wide dropdowns alongside compact action selectors).
- **Viewports & Content Collapses:** Fixing layout distortions where multi-column inputs overflow or clip text zones on standard tablet dimensions.
- **State Syncing Hurdles:** Preventing data context drops when converting structured JSON strings into active text input values.

Solutions Applied

How Challenges Were Resolved

- **CSS Grid Layout Subsplits:** Applied discrete column tracks on target components to keep nested input fields perfectly aligned.
- **Unified Media Resets:** Configured a single block breakpoint (`@media(max-width:1100px)`) that safely collapses rows down to full-width blocks on mobile views.
- **Robust DOM Binding Verification:** Built clear checks (`if(form)`) to confirm DOM nodes load completely before executing form actions.

Code Walkthrough

State Ingestion & Validation Module:

```
function loginUser() {
  let appNo = document.getElementById("appNo").value;
  let student = localStorage.getItem(appNo);

  if (student) {
    sessionStorage.setItem("currentPatient", appNo);
    window.location.href = "registration.html";
  } else {
    alert("Registration Not Found");
  }
}
```

Multi-Column Workspace Splitting:

```
.form-grid {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 40px;
}

@media(max-width:1100px) {
  .form-grid { grid-template-columns: 1fr; }
  .double-input, .triple-input {
    grid-template-columns: 1fr;
  }
}
```

Testing & Validation

Testing Evidence & Verification Steps

- **Gateway Guard Interceptions:** Checked that typing an incorrect token correctly stops navigation and displays a clear alert box.
- **Form Fluidity Validations:** Resized window frames down to 1000px wide to confirm columns stack correctly without horizontal scrollbars appearing.
- **State Restoration Verification:** Verified that active profile variables reload cleanly from local memory straight into their target form elements.

Results & Achievements

Core Deliverables Realized

- **Seamless Clinical Entry Gateway:** Developed an interconnected interface framework that links search tokens directly to intake screens.
- **Highly Functional Intake UI:** Designed a beautiful, comprehensive multi-column form structure that remains perfectly stable during layout changes.

Engineering Milestones

- Isolated visual styling layouts entirely into structured stylesheets.
- Mastered reliable cross-page state caching using browser memory features.

Key Learning

Technical & Professional Growth

- **Enterprise Forms Architecture:** Learned how to structure intricate layouts across multi-tier fields without breaking visual consistency.
- **Client State Life Cycle Controls:** Mastered the mechanics of saving, validating, and pulling transient object structures across browser view changes.
- **Industrial Documentation Competence:** Transformed operational application code into well-structured, production-focused presentation layers ready for senior design reviews.